

AIC LAB

Experiment 6: AI-Based Code Completion — Working with Suggestions for Classes, Loops, Conditionals

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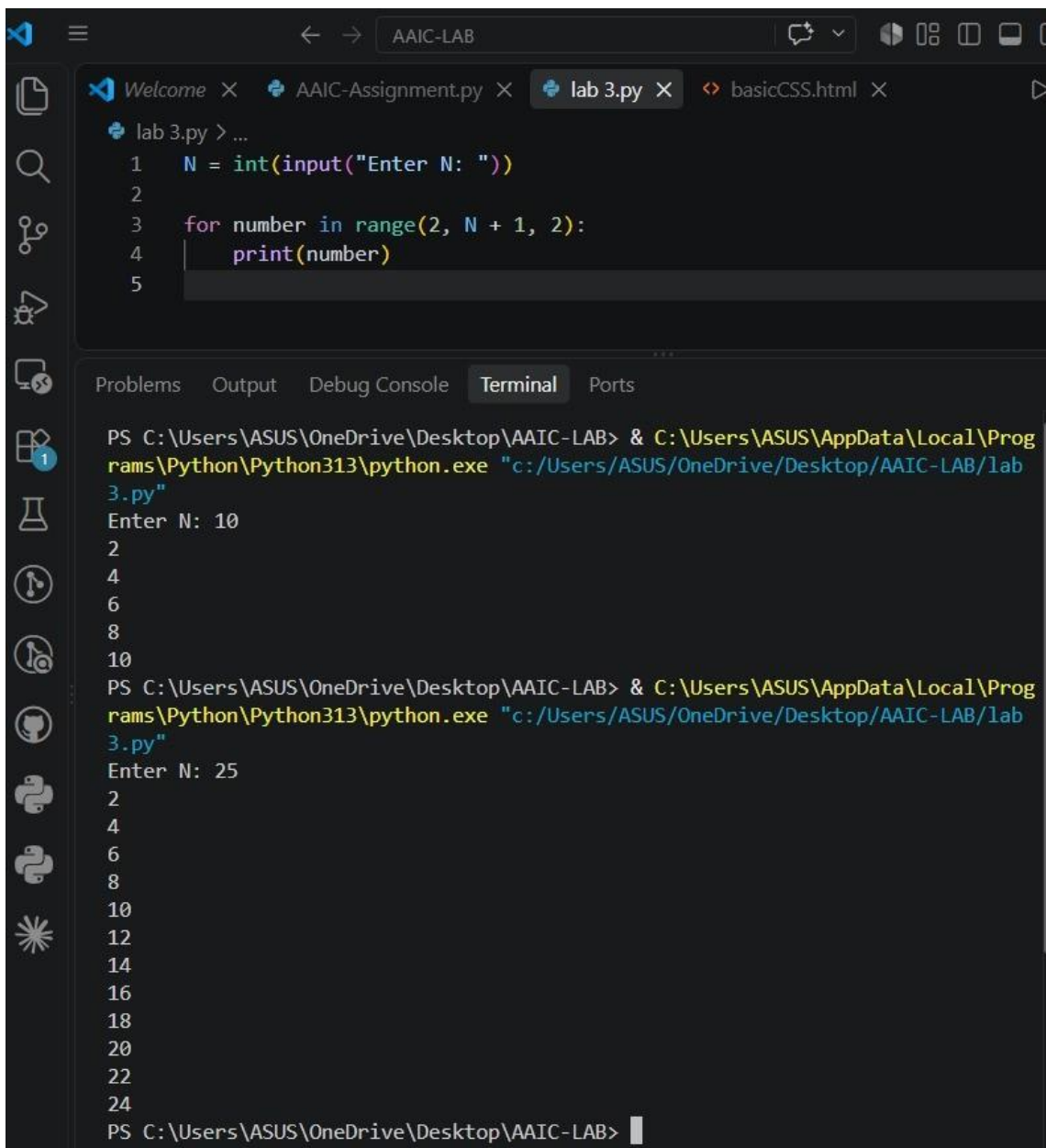
Roll No :2503A52692

Batch No : 24

Task Description #1 — AI-Based Code Completion for Loops

Prompt: "Generate Python code to print all even numbers between 1 and N using a loop."

#code & #output



The image shows a Visual Studio Code window with the title bar 'AAIC-LAB'. The editor has four tabs: 'Welcome', 'AAIC-Assignment.py', 'lab 3.py', and 'basicCSS.html'. The 'lab 3.py' tab is active, displaying the following Python code:

```
lab 3.py > ...
1  N = int(input("Enter N: "))
2
3  for number in range(2, N + 1, 2):
4      print(number)
5
```

Below the editor, the 'Terminal' panel is open, showing the execution of the script. The command prompt shows the command to run 'lab 3.py' from the directory 'C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB'. The output shows the program running twice: first with N=10, printing even numbers from 2 to 10, and then with N=25, printing even numbers from 2 to 24.

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & C:\Users\ASUS\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/ASUS/OneDrive/Desktop/AAIC-LAB/lab 3.py"
Enter N: 10
2
4
6
8
10
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & C:\Users\ASUS\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/ASUS/OneDrive/Desktop/AAIC-LAB/lab 3.py"
Enter N: 25
2
4
6
8
10
12
14
16
18
20
22
24
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```

Task Description #2 — AI-Based Code Completion for Loop with Conditionals

Prompt: "Generate Python code to count how many numbers in a list are even and odd."

#code & #output

```
lab 3.py 7 ...
1  numbers = list(map(int, input("Enter numbers separated by spaces: ").split()))
2
3  even_count = sum(number % 2 == 0 for number in numbers)
4  odd_count = sum(number % 2 != 0 for number in numbers)
5
6  print("Even numbers:", even_count)
7  print("Odd numbers:", odd_count)
8  |

Problems Output Debug Console Terminal Ports
+ ~
Missing argument in parameter list.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecord
Exception
+ FullyQualifiedErrorId : MissingTypeName

PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & C:\Users\ASUS\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/ASUS/OneDrive/Desktop/AAIC-LAB/lab 3.py"
Enter numbers separated by spaces: 10 15 22 33 48 51 60 7
Even numbers: 4
Odd numbers: 4
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & C:\Users\ASUS\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/ASUS/OneDrive/Desktop/AAIC-LAB/lab 3.py"
Enter numbers separated by spaces: 1 14 10 96 02 45
Even numbers: 4
Odd numbers: 2
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> |
```

Task Description #3 — AI-Based Code Completion for Class Attributes Validation

Prompt: "Generate a Python class User that validates age and email using conditional statements."

#code & #output

```
lab3.py
1  class User:
2      def __init__(self, name, age, email):
3          self.name = name
4          self.age = age
5          self.email = email
6          self.is_valid = self.validate()
7
8      def validate(self):
9          if self.age < 0 or self.age > 120:
10             print("Invalid age: age must be between 0 and 120")
11             return False
12          if "@" not in self.email or "." not in self.email:
13             print("Invalid email: must contain '@' and '.'")
14             return False
15          print(f"{self.name} is valid. Age: {self.age}, Email: {self.email}")
16          return True
17
18
19 name = input("Enter name: ")
20 age = int(input("Enter age: "))
21 email = input("Enter email: ")
22 user = User(name, age, email)
```

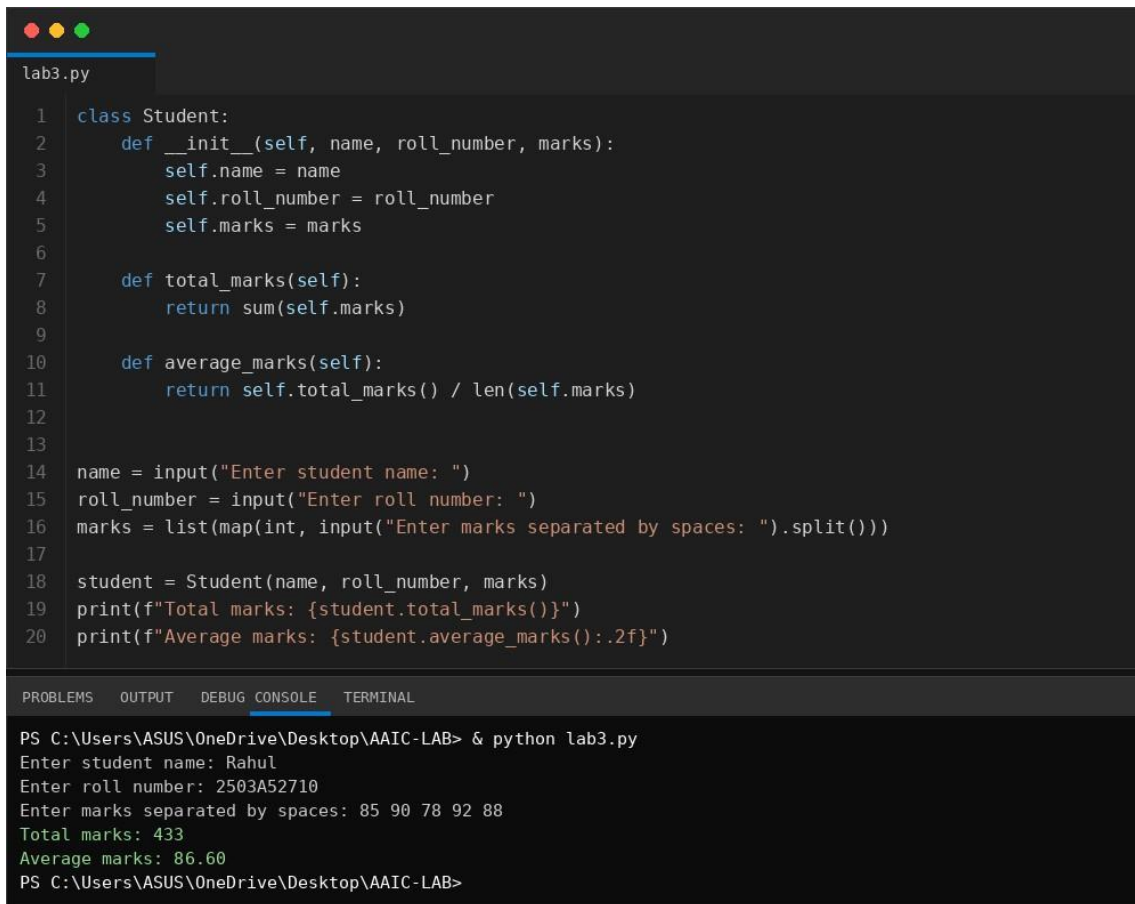
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter name: Vinay
Enter age: 25
Enter email: vinay@gmail.com
Vinay is valid. Age: 25, Email: vinay@gmail.com
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter name: Sam
Enter age: 150
Enter email: sam@gmail.com
Invalid age: age must be between 0 and 120
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter name: Anu
Enter age: 30
Enter email: anugmail.com
Invalid email: must contain '@' and '.'
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```

Task Description #4 — AI-Based Code Completion for Classes

Prompt: "Generate a Python class Student with attributes (name, roll number, marks) and methods to calculate total and average marks."

#code & #output



```
lab3.py
1 class Student:
2     def __init__(self, name, roll_number, marks):
3         self.name = name
4         self.roll_number = roll_number
5         self.marks = marks
6
7     def total_marks(self):
8         return sum(self.marks)
9
10    def average_marks(self):
11        return self.total_marks() / len(self.marks)
12
13
14    name = input("Enter student name: ")
15    roll_number = input("Enter roll number: ")
16    marks = list(map(int, input("Enter marks separated by spaces: ").split()))
17
18    student = Student(name, roll_number, marks)
19    print(f"Total marks: {student.total_marks()}")
20    print(f"Average marks: {student.average_marks():.2f}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter student name: Rahul
Enter roll number: 2503A52710
Enter marks separated by spaces: 85 90 78 92 88
Total marks: 433
Average marks: 86.60
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```

Task Description #5 — AI-Assisted Code Completion Review

Prompt: "Generate a Python program for a simple bank account system using class, loops, and conditional statements."

#code & #output

lab3.py

```
1 class BankAccount:
2     def __init__(self, owner, balance=0):
3         self.owner = owner
4         self.balance = balance
5
6     def deposit(self, amount):
7         if amount <= 0:
8             print("Deposit amount must be positive.")
9         else:
10            self.balance += amount
11            print(f"Deposited {amount}. New balance: {self.balance}")
12
13    def withdraw(self, amount):
14        if amount <= 0:
15            print("Withdrawal amount must be positive.")
16        elif amount > self.balance:
17            print("Insufficient balance.")
18        else:
19            self.balance -= amount
20            print(f"Withdrew {amount}. New balance: {self.balance}")
21
22    def check_balance(self):
23        print(f"Current balance: {self.balance}")
24
25
26 name = input("Enter account holder name: ")
27 account = BankAccount(name)
28
29 while True:
30     print("1. Deposit 2. Withdraw 3. Check Balance 4. Exit")
31     choice = input("Enter choice: ")
32     if choice == "1":
33         amount = float(input("Enter amount to deposit: "))
34         account.deposit(amount)
35     elif choice == "2":
36         amount = float(input("Enter amount to withdraw: "))
37         account.withdraw(amount)
38     elif choice == "3":
39         account.check_balance()
40     elif choice == "4":
41         print("Exiting... Thank you!")
42         break
43     else:
44         print("Invalid choice. Please try again.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> & python lab3.py
Enter account holder name: Priya
1. Deposit 2. Withdraw 3. Check Balance 4. Exit
Enter choice: 1
Enter amount to deposit: 5000
Deposited 5000.0. New balance: 5000.0
1. Deposit 2. Withdraw 3. Check Balance 4. Exit
Enter choice: 3
Current balance: 5000.0
1. Deposit 2. Withdraw 3. Check Balance 4. Exit
Enter choice: 2
Enter amount to withdraw: 7000
Insufficient balance.
1. Deposit 2. Withdraw 3. Check Balance 4. Exit
Enter choice: 2
Enter amount to withdraw: 2000
Withdrew 2000.0. New balance: 3000.0
1. Deposit 2. Withdraw 3. Check Balance 4. Exit
Enter choice: 4
Exiting... Thank you!
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB>
```